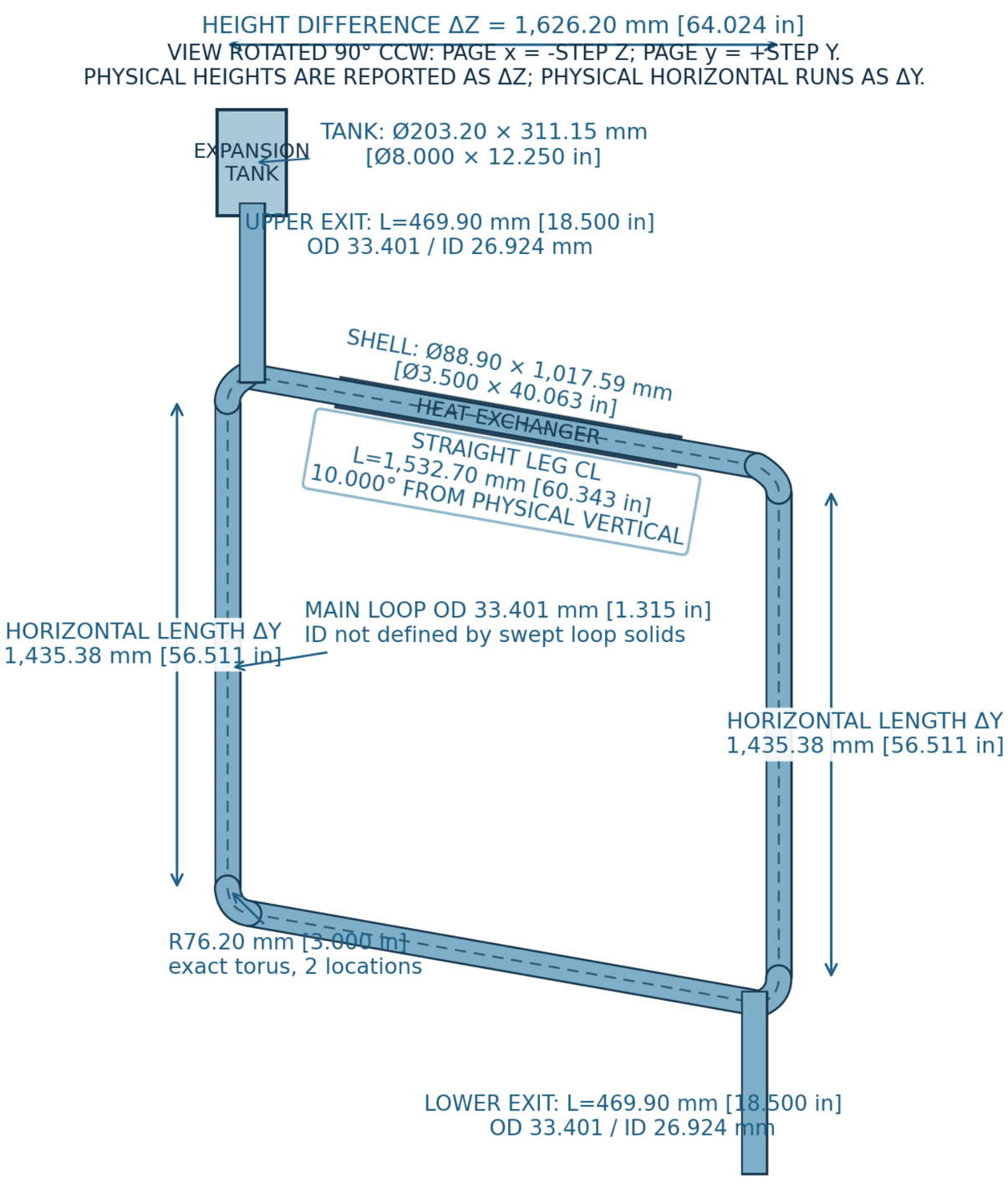
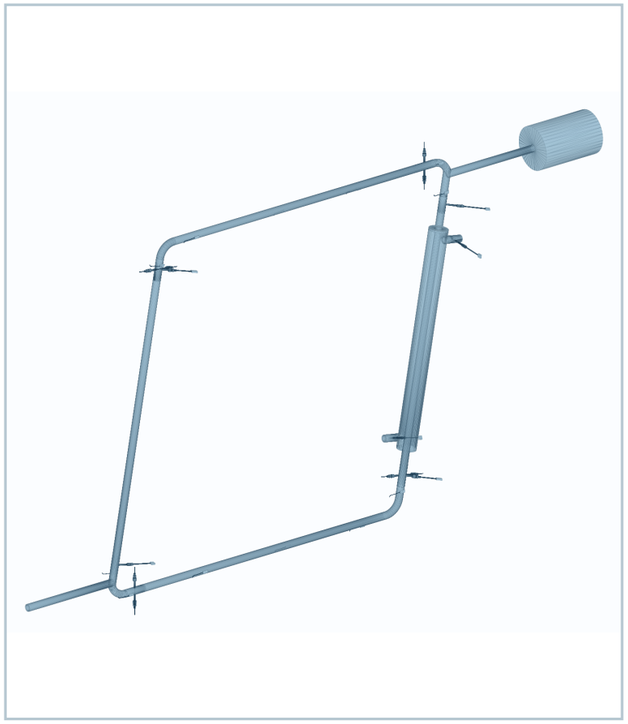
VCU GENERAL ARRANGEMENT AND PRIMARY DIMENSIONS

VCU LOOP WITH OPTICS AND HEAT EXCHANGER - SOURCE MODEL REVISION 8



CAD ASSEMBLY - ISOMETRIC REFERENCE



MODEL ENVELOPE (INCLUDING SENSORS)

Y span	3,101.55 mm [122.108 in]
Z span	1,856.80 mm [73.103 in]
X depth	317.72 mm [12.509 in]

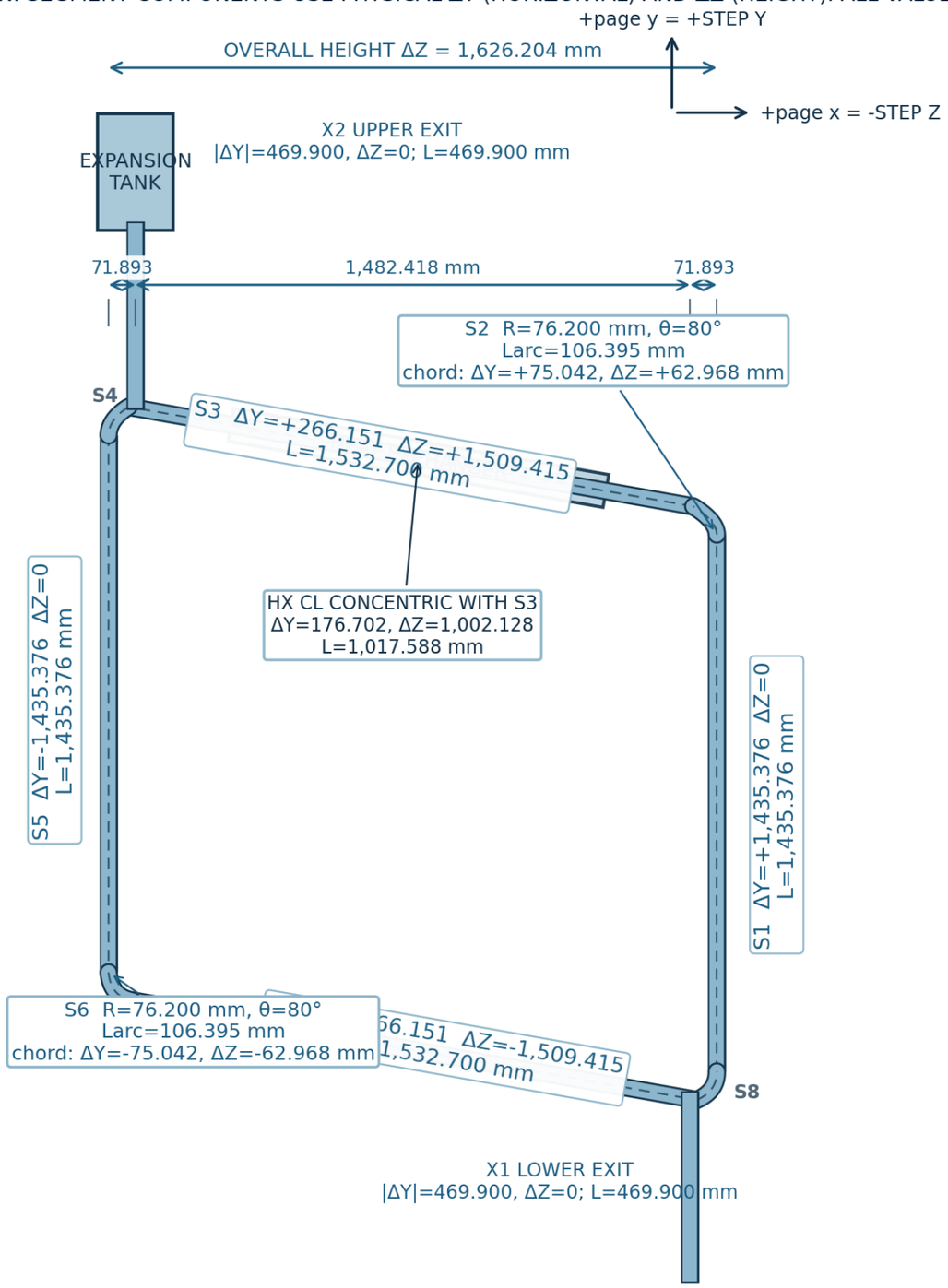
FABRICATION LIMITATION

The STEP assembly contains geometry but no drawing tolerances, weld symbols, material callouts, inspection requirements, or authoritative sensor channel names. This blueprint is suitable for engineering review and experimental planning, not fabrication release.

SOURCE MODEL	DRAWING STATUS	UNITS / SCALE	REV / SHEET
UTDTloopWithOpticsAndHXRev8.STEP	PRELIMINARY - FOR ENGINEERING REVIEW	mm [in] / NTS	C / 1 OF 5

CENTERLINE AND SEGMENT DIMENSIONS

VIEW ROTATED 90° CCW. SEGMENT COMPONENTS USE PHYSICAL ΔY (HORIZONTAL) AND ΔZ (HEIGHT). ALL VALUES IN mm.



SEGMENT SCHEDULE - LOCAL SIDE VIEW

SEG	ΔY mm	ΔZ mm	L / Lchord mm
S1	+1,435.376	0.000	1,435.376
S2	+75.042	+62.968	106.395 arc
S3	+266.151	+1,509.415	1,532.700
S4	-76.173	+53.821	93.269 chord
S5	-1,435.376	0.000	1,435.376
S6	-75.042	-62.968	106.395 arc
S7	-266.151	-1,509.415	1,532.700
S8	+76.173	-53.821	93.269 chord
X1	-469.900	0.000	469.900
X2	+469.900	0.000	469.900
HX	+176.702	+1,002.128	1,017.588
T1	+311.150	0.000	311.150

HORIZONTAL-PIPE CENTERLINE CHAIN

Plot abbreviations: BR bottom run; LE lower exit; UE upper exit; TR top run.

Horizontal pipe-axis chain:

- Bottom run → lower exit: 71.893 mm
- Lower exit → upper exit: 1,482.418 mm
- Upper exit → top run: 71.893 mm
- Bottom run → top run: 1,626.204 mm

Heat-exchanger shell axis and S3/right-leg axis are concentric in the STEP geometry: normal offset = 0 mm.

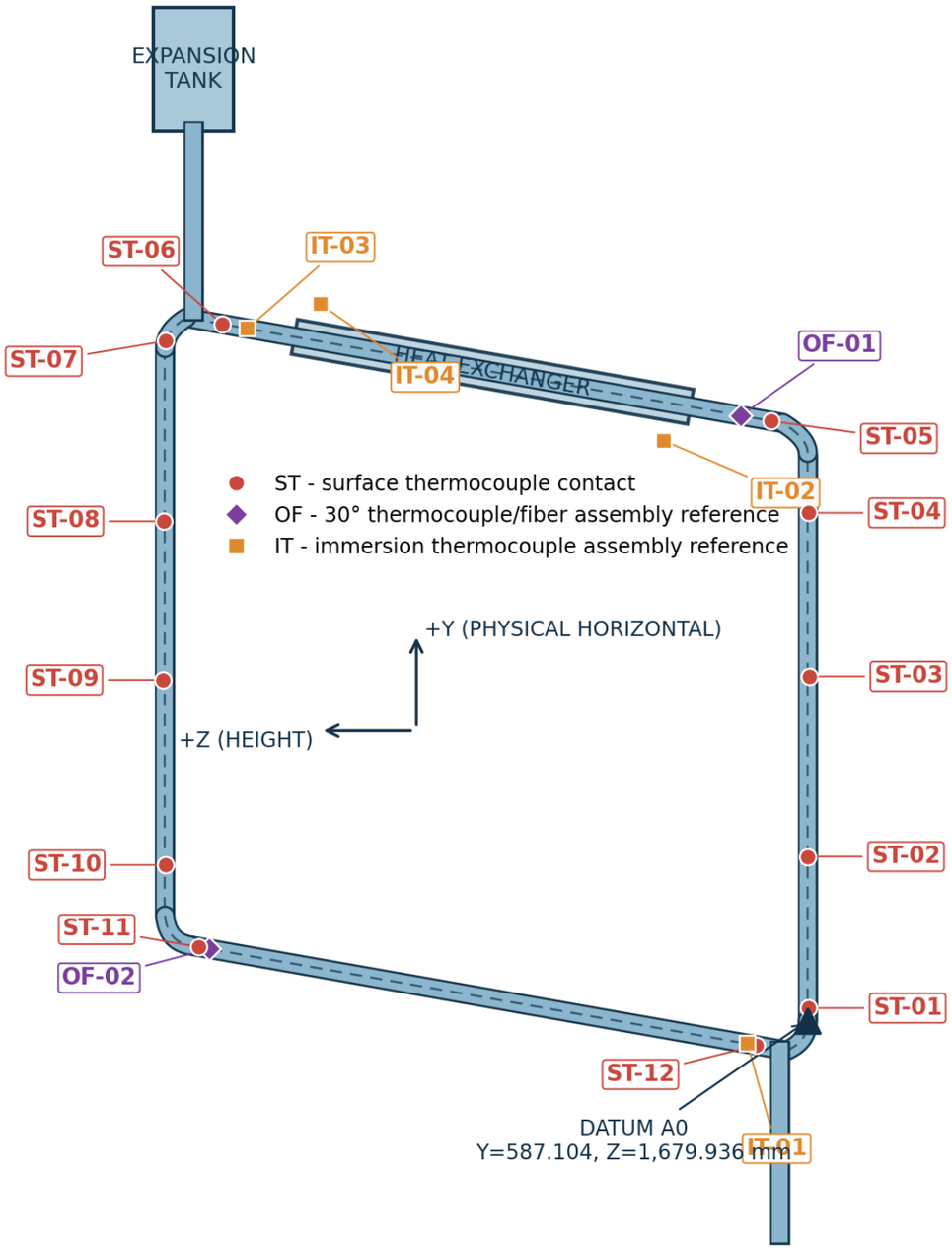
The view is rotated 90° CCW. Segment components remain physical ΔY (horizontal) and ΔZ (height).

S4/S8 values are endpoint chords, not arc lengths.

SOURCE MODEL	DRAWING STATUS	UNITS / SCALE	REV / SHEET
UTDTloopWithOpticsAndHXRev8.STEP	PRELIMINARY - FOR ENGINEERING REVIEW	mm [in] / NTS	C / 2 OF 5

SENSOR LOCATION PLAN

ROTATED VIEW: EXPANSION TANK UPPER LEFT



SENSOR OFFSETS FROM DATUM A0

ΔY = physical horizontal location; ΔZ = height difference. Values in mm.

ID	STEP	ΔY horiz.	ΔZ height
IT-01	NAUO23	-58.904	+152.139
IT-02	NAUO22	+1,465.989	+364.821
IT-03	NAUO24	+1,749.221	+1,417.624
IT-04	NAUO9	+1,810.255	+1,232.649
OF-01	NAUO7	+1,529.122	+169.286
OF-02	NAUO8	+181.244	+1,513.969
ST-01	NAUO20	+31.381	-3.043
ST-02	NAUO19	+414.476	+1.165
ST-03	NAUO18	+870.254	-4.871
ST-04	NAUO17	+1,283.361	-1.016
ST-05	NAUO16	+1,515.430	+92.561
ST-06	NAUO15	+1,760.954	+1,482.217
ST-07	NAUO14	+1,717.814	+1,624.572
ST-08	NAUO11	+1,262.085	+1,628.821
ST-09	NAUO12	+861.207	+1,631.086
ST-10	NAUO13	+393.110	+1,625.476
ST-11	NAUO10	+186.192	+1,541.874
ST-12	NAUO21	-62.774	+129.427

Drawing-local IDs remain traceable to STEP instance names; confirm experiment channel mapping before release.

ST points are geometric contacts with the modeled loop outer surface. OF/IT points are assembly placement references at the modeled mount/main-pipe solid centroid; sensing-junction insertion depth is not encoded as a named datum in the STEP file.

SOURCE MODEL	DRAWING STATUS	UNITS / SCALE	REV / SHEET
UTDTloopWithOpticsAndHXRev8.STEP	PRELIMINARY - FOR ENGINEERING REVIEW	mm [in] / NTS	C / 3 OF 5

SENSOR COORDINATE SCHEDULE

VCU LOOP WITH OPTICS AND HEAT EXCHANGER - SOURCE MODEL REVISION 8

Datum A0 = lower-straight left tangent in the loop center plane: X = 629.807, Y = 587.104, Z = 1,679.936 mm. Bracketed values are inches.

ID	STEP	REFERENCE TYPE	GLOBAL X mm [in]	GLOBAL Y mm [in]	GLOBAL Z mm [in]	ΔX mm [in]	ΔY mm [in]	ΔZ mm [in]
IT-01	NAUO23	Immersion TC assembly	629.805 [24.7955]	528.200 [20.7953]	1832.075 [72.1289]	-0.002 [-0.0001]	-58.904 [-2.3191]	152.139 [5.9897]
IT-02	NAUO22	Immersion TC assembly	628.995 [24.7636]	2053.093 [80.8304]	2044.757 [80.5023]	-0.812 [-0.0320]	1465.989 [57.7161]	364.821 [14.3630]
IT-03	NAUO24	Immersion TC assembly	629.776 [24.7943]	2336.325 [91.9813]	3097.560 [121.9512]	-0.031 [-0.0012]	1749.221 [68.8670]	1417.624 [55.8120]
IT-04	NAUO9	Immersion TC assembly	630.458 [24.8212]	2397.359 [94.3842]	2912.585 [114.6687]	0.651 [0.0256]	1810.255 [71.2699]	1232.649 [48.5295]
OF-01	NAUO7	30° TC/fiber assembly	630.052 [24.8052]	2116.226 [83.3160]	1849.222 [72.8040]	0.245 [0.0096]	1529.122 [60.2016]	169.286 [6.6648]
OF-02	NAUO8	30° TC/fiber assembly	629.786 [24.7947]	768.348 [30.2499]	3193.905 [125.7443]	-0.021 [-0.0008]	181.244 [7.1356]	1513.969 [59.6051]
ST-01	NAUO20	Surface TC contact	646.248 [25.4428]	618.485 [24.3498]	1676.893 [66.0194]	16.441 [0.6473]	31.381 [1.2355]	-3.043 [-0.1198]
ST-02	NAUO19	Surface TC contact	646.552 [25.4548]	1001.580 [39.4323]	1681.101 [66.1851]	16.745 [0.6593]	414.476 [16.3180]	1.165 [0.0459]
ST-03	NAUO18	Surface TC contact	645.704 [25.4214]	1457.358 [57.3763]	1675.065 [65.9474]	15.897 [0.6259]	870.254 [34.2620]	-4.871 [-0.1918]
ST-04	NAUO17	Surface TC contact	646.711 [25.4611]	1870.465 [73.6403]	1678.920 [66.0992]	16.904 [0.6655]	1283.361 [50.5260]	-1.016 [-0.0400]
ST-05	NAUO16	Surface TC contact	646.778 [25.4637]	2102.534 [82.7769]	1772.497 [69.7833]	16.971 [0.6681]	1515.430 [59.6626]	92.561 [3.6441]
ST-06	NAUO15	Surface TC contact	646.505 [25.4529]	2348.058 [92.4432]	3162.153 [124.4942]	16.698 [0.6574]	1760.954 [69.3289]	1482.217 [58.3550]
ST-07	NAUO14	Surface TC contact	646.470 [25.4516]	2304.918 [90.7448]	3304.508 [130.0987]	16.663 [0.6560]	1717.814 [67.6305]	1624.572 [63.9595]
ST-08	NAUO11	Surface TC contact	646.044 [25.4348]	1849.189 [72.8027]	3308.757 [130.2660]	16.237 [0.6393]	1262.085 [49.6884]	1628.821 [64.1268]
ST-09	NAUO12	Surface TC contact	645.778 [25.4243]	1448.311 [57.0201]	3311.022 [130.3552]	15.971 [0.6288]	861.207 [33.9058]	1631.086 [64.2160]
ST-10	NAUO13	Surface TC contact	646.491 [25.4524]	980.214 [38.5911]	3305.412 [130.1343]	16.684 [0.6569]	393.110 [15.4768]	1625.476 [63.9951]
ST-11	NAUO10	Surface TC contact	646.507 [25.4530]	773.296 [30.4447]	3221.810 [126.8429]	16.700 [0.6575]	186.192 [7.3304]	1541.874 [60.7037]
ST-12	NAUO21	Surface TC contact	646.507 [25.4530]	524.330 [20.6429]	1809.363 [71.2347]	16.700 [0.6575]	-62.774 [-2.4714]	129.427 [5.0955]

Model identifiers are preserved for traceability. Proposed ST/OF/IT tags are drawing-local identifiers and must be cross-referenced to the experiment data-acquisition channel list before release.

SOURCE MODEL	DRAWING STATUS	UNITS / SCALE	REV / SHEET
UTDTloopWithOpticsAndHXRev8.STEP	PRELIMINARY - FOR ENGINEERING REVIEW	mm [in] / NTS	C / 4 OF 5

COMPONENT DIMENSION SCHEDULE AND MODEL NOTES

ASSEMBLY-LEVEL DIMENSIONS EXTRACTED FROM THE STEP GEOMETRY

FEATURE	QTY	MODELED DIMENSION	BASIS / COMMENT
Main loop outside diameter	1	33.401 mm [1.315 in]	Exact cylindrical outer surface. Main swept-loop solids do not define an internal flow-passage diameter.
Top straight centerline	1	1,435.376 mm [56.511 in]	Cylinder-axis endpoints; top horizontal run.
Bottom straight centerline	1	1,435.376 mm [56.511 in]	Cylinder-axis endpoints; bottom horizontal run.
Straight sloping leg	2	1,532.700 mm [60.343 in]	Combined collinear cylinders per leg; 10.000° from global vertical.
Top/bottom centerline separation	1	1,626.204 mm [64.024 in]	Difference between horizontal-run cylinder axes.
Exact torus bend centerline radius	2	76.200 mm [3.000 in]	Exact toroidal faces found at upper-left and lower-right. Other corner joins are swept/joined model geometry.
Horizontal exit tube	2	L 469.900; OD 33.401; ID 26.924 mm	Exact cylindrical surfaces. [L 18.500; OD 1.315; ID 1.060 in].
Expansion tank	1	OD 203.200 × L 311.150 mm	Exact exterior cylinder. [OD 8.000 × L 12.250 in].
Heat-exchanger shell	1	OD 88.900 × L 1,017.588 mm	Exact exterior cylinder, aligned with 10° leg. [OD 3.500 × L 40.063 in].
Heat-exchanger shell bore	1	ID 77.927 mm [3.068 in]	Exact interior cylindrical face of shell component.
Surface thermocouples	12	See sensor coordinate schedule	Reference is closest geometric contact point on main loop exterior.
30° TC/fiber assemblies	2	See sensor coordinate schedule	Reference is centroid of modeled local main-pipe member.
Immersion TC assemblies	4	See sensor coordinate schedule	Reference is centroid of modeled mounting/main-pipe body.

INTERPRETATION AND RELEASE NOTES

1. Coordinate values are reported to 0.001 mm to preserve CAD traceability; this precision does not imply manufacturing accuracy.
2. Global X is normal to the primary loop plane. Global Y is horizontal in the side view. Global Z is vertical.
3. No functional channel names (for example TP3, TP4, or TW1) are attached to sensor instances in the STEP assembly. The drawing-local IDs therefore retain the STEP NAUO instance in the schedule.
4. Purchased-part internal features, threads, compression fittings, wire routing, weld preparation, and insertion depth are not fully dimensioned here. Use the manufacturer drawing and the experiment installation record for release.
5. Confirm the modeled main-loop internal diameter and all weld/tolerance requirements before using this drawing for thermal-hydraulic volume, pressure-drop, or fabrication calculations.